Hospital Readmission Risk Predictor

Preprocessing & Feature Engineering Report

A Phased MLOps Build

Prepared for: Hospital Administration & Care Management Stakeholders

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GitHub Repository:

https://github.com/AngelineSetiawan/AngelineSetiawan-hospital-readmission-risk-predictor

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1. Background & Question

This report is the second deliverable in the Hospital Readmission Risk Predictor project, following the Exploratory Data Analysis (EDA) Report submitted on July 21, 2026. Where that report characterized the raw data, this report documents what was actually done to the data before modeling: which rows and columns were changed, dropped, or engineered, and why each decision was made. The research question, hypothesis, and prediction carried over from the proposal are restated below for continuity, since no changes were made to any of them during this phase.

Research Question

Can a patient's risk of being readmitted to the hospital within 30 days of discharge be predicted from routinely captured clinical encounter data, and can that prediction be delivered through a live, self-updating system rather than a one-time report?

Hypothesis & Prediction

Patients with more complex or intensive care histories, meaning longer hospital stays, more prior inpatient or emergency visits, and a higher number of medications, are more likely to be readmitted within 30 days, because these factors reflect greater underlying disease severity and more complicated post-discharge care needs. The corresponding prediction is that a supervised classification model trained on the resulting features will outperform a naive majority-class baseline on AUC-ROC and recall for the readmitted class, and that utilization-related features, specifically prior inpatient visits, time in hospital, and number of medications, will rank among the model's most important predictors.

How This Phase Connects to the EDA Findings

The preprocessing plan described below is not a generic checklist. It is a direct response to three findings from the EDA Report. First, the class imbalance documented in that report (11.2% of encounters resulted in a 30-day readmission) is preserved rather than altered here, since resampling decisions are deferred to the modeling phase where they can be evaluated against a held-out set rather than baked into the data itself. Second, the 1,652 encounters ending in patient death, flagged in the EDA report as guaranteed negative-class examples with no bearing on clinical risk, are removed in this phase to prevent that structural bias from reaching the model. Third, the hidden missingness identified in the admission and discharge label columns, and the informative, non-random missingness in the glucose and A1C lab fields, are handled here according to the mechanism behind each: encoded as an explicit category where missingness itself may carry signal, rather than imputed or discarded.

2. Methods

2.1 Restatement of the Preprocessing Plan and Key EDA Findings

The original preprocessing plan, proposed before any exploratory analysis, called for converting raw inpatient diabetic encounters from 130 U.S. hospitals into a model-ready table with minimal assumptions about which fields would matter. The EDA Report revised that plan in three concrete ways that this phase now implements.

- **Missingness is not random.** Glucose and HbA1c lab fields are missing because a physician chose not to order the test, not because the record is incomplete, so these are encoded as testing indicators rather than imputed. Patient weight, by contrast, is missing for roughly 97% of encounters with no recoverable pattern behind it, so it is dropped outright.

- **In-hospital deaths bias the target.** 1,652 encounters end in patient death. Every one of these is coded "not readmitted" for a reason that has nothing to do with clinical risk, since a deceased patient cannot return to the hospital. Leaving them in would artificially inflate the negative class and understate the model's true difficulty.

- **Repeated patients violate independence.** Over 23% of unique patients in the cleaned data have more than one encounter. Treating each encounter as an independent observation would let a single patient's history leak across the train and test partitions and would distort feature-importance estimates toward frequently readmitted patients. This required an explicit patient-level aggregation strategy, described in Section 2.3.

2.2 Preprocessing Methods and Rationale

Preprocessing was implemented as a single, deterministic, and reproducible transformation pipeline, executed in a fixed order and version-controlled in the project's GitHub repository so that any analyst can regenerate the cleaned table directly from the original UCI download. Each step below states what was done and, more importantly, why it was done, since a preprocessing decision that cannot be justified to a clinical stakeholder is a decision that should not be made silently.

- **Label mapping:** The numeric codes for admission type, discharge disposition, and admission source were joined against the UCI reference table to produce human-readable label columns, while the original numeric codes were preserved unchanged. This keeps the data usable for both a non-technical audit of the cleaning logic and for numeric modeling, without forcing a choice between the two.
- **Null standardization:** The dataset's "?" placeholder was converted to a proper null across every column. This is a prerequisite for accurate missingness counting and for any downstream imputation or encoding logic to behave predictably.
- **Target derivation:** The native three-class readmission label was collapsed into a binary target, readmitted_30d, that flags only readmissions occurring within 30 days. This aligns the modeling target directly with the outcome measured by CMS under the Hospital Readmissions Reduction Program, which is the outcome the eventual model needs to predict to be operationally useful. The original three-class label was then removed from the feature matrix so it cannot leak into the model as a disguised version of the target.

- **Mortality exclusion:** The 1,652 encounters coded with an in-hospital-death discharge disposition were removed entirely, consistent with the original dataset publication's own handling and with the reasoning in Section 2.1. This reduced the working dataset from 101,766 to 100,114 encounters.

- **Weight removal:** The weight column was dropped completely. At 96.86% missingness, no imputation strategy would produce values reliable enough to trust in a clinical risk score, and retaining a near-empty column adds noise without adding signal.

- **Testing indicators:** The max_glu_serum and A1Cresult fields were converted into binary tested and not-tested indicators rather than having their missing values filled in. Because more than 90% of glucose tests and roughly 80% of A1C tests were not ordered at the encounter, the fact that a test was or was not ordered is itself a piece of clinical information, likely reflecting physician judgment about the patient's presentation, and that signal would be lost under a conventional imputation approach.

- **Explicit missing category for demographic and administrative fields:** For race, medical_specialty, and payer_code, missing values were assigned their own explicit category rather than being dropped or imputed to the most common value. This preserves every row for modeling and allows the model to learn whether missingness itself, for example an unrecorded payer code, carries any administrative or systemic pattern worth surfacing to a care team, rather than assuming missingness is uninformative.

- **Unknown-category consolidation:** The literal placeholder strings inherited from the UCI reference table, such as "NULL," "Not Available," and "Not Mapped," were consolidated into a single explicit "unknown" category per field. Leaving these as separate categories would fragment an already sparse signal across near-duplicate labels for no analytical benefit.

Imputation Alternatives Considered and Rejected

Because the EDA Report's missingness audit surfaced nine columns with meaningful missing data, standard imputation approaches, mean or median substitution for continuous fields and mode substitution for categorical fields, were evaluated as a baseline alternative before any field-specific decision was finalized. They were rejected as a default strategy for the reasons below, which is why the field-specific handling in Table 2 (Section 3.2) varies by field rather than applying one rule uniformly.

- **Mean or median imputation for max_glu_serum and A1Cresult** was rejected because the missingness in these two fields is not random. It reflects a clinical decision not to order a test, and filling that gap with a typical lab value would fabricate a result the physician never recorded, misrepresenting the encounter to any downstream model or reviewer.
- **Mode imputation for medical_specialty and payer_code** was rejected because both fields are high-cardinality and administratively rather than clinically driven. Assigning every missing record to the single most common specialty or payer would silently overweight that category and could mask a genuine administrative pattern, such as a class of encounters that is systematically unbilled or unassigned, that is worth preserving as its own signal.
- **Mode imputation for race** was rejected on a narrower but firmer basis: race is a protected demographic attribute, and silently reassigning a patient's unrecorded race to

the majority category (Caucasian, at 74.78% of encounters) would introduce a systematic, direction-biased distortion into a field this project treats as sensitive. An explicit missing category avoids fabricating a demographic label the source record does not support.

- **Regression or k-nearest-neighbor imputation** was considered as a more sophisticated alternative for the two lab-result fields but was set aside for the same reason as mean imputation: any model-based estimate of a lab result the physician chose not to order still fabricates a value, and a more sophisticated fabrication is not a more defensible one for a field where missingness itself is the signal of interest.

In every case, the guiding test was whether imputation would preserve or destroy the clinical meaning of the missing value. Where missingness plausibly reflects a decision (ordering a test, recording a specialty), it is encoded as its own category or indicator rather than filled in. This is an assumption, not a finding: the evidence behind it is the data dictionary's description of how each field is populated, not a formal missingness test, such as Little's MCAR test or a comparison of outcome rates between missing and non-missing groups, run against this dataset. No such test was performed in this phase. What the available evidence does support is a narrower, defensible claim: for the nine fields with meaningful missing data, there is a plausible, documented mechanism, a clinical or administrative decision not to capture the value, that would make conventional mean, median, or mode imputation likely to fabricate or misrepresent that value. That is enough to justify not imputing these fields by default. It is not enough to assert that the missingness is formally non-random, and this report does not make that stronger claim.

2.3 Feature Engineering Methods and Rationale

Feature engineering focused on two problems the EDA Report flagged as high priority: diagnosis codes that are too sparse to use directly, and a temporal structure that has to be respected to avoid leakage.

- **Diagnosis code grouping:** diag_1, diag_2, and diag_3 each contain hundreds of distinct ICD-9 codes. One-hot encoding them directly would create a feature space large relative to the number of available rows and would fragment clinically related conditions into hundreds of near-empty columns. Each code was instead mapped into one of nine clinically meaningful categories (circulatory, respiratory, digestive, diabetes, genitourinary, musculoskeletal, injury, neoplasms, and other), following standard ICD-9 chapter groupings. This keeps the diagnosis signal usable by a model while remaining interpretable to a clinical reviewer, who can reason about a "circulatory diagnosis" in a way they cannot reason about ICD-9 code 414.01.

- **Chronology proxy:** Because the public release of this dataset does not include exact calendar timestamps, encounter_id was used to construct an ordinal time index (encounter_order), consistent with the time-aware validation strategy set out in the original proposal. This rests on an explicit assumption that this report does not treat lightly: that encounter_id increases monotonically with the actual calendar time of the encounter. That assumption is not verified against ground truth in this phase; it is not, and cannot be, tested directly, since no calendar timestamp exists anywhere in the public release to check it against. The basis for the assumption is the UCI data dictionary's description of encounter_id as a sequentially assigned identifier, not an empirical demonstration that sorting by it reproduces true chronological order. It is entirely possible that encounter IDs were assigned in a way that is only loosely correlated with calendar time, for example if multiple hospitals' records were batch-loaded, backfilled, or merged into the source system out of strict time order. If that is the case, encounter_order is a noisy proxy for time rather than a clean one, and everything downstream that depends on it, the first-encounter snapshot in this section and the chronological train/test split in Section 3.4, inherits that same uncertainty. This report proceeds on the assumption because it is the best ordering signal available in the public data and because the original proposal committed to a time-aware validation strategy that requires some ordering variable to implement, not because the assumption has been independently confirmed. The practical consequences of this assumption, and what would need to be true for it to fail, are discussed further in Section 3.4 and revisited in the Limitations in Section 4.3.

- **Patient-level snapshot:** To resolve the independence violation described in Section 2.1 without discarding the project's ability to use prior-utilization counts, each patient's first eligible encounter (earliest by encounter_order) was retained as that patient's single analytical row, and all later encounters for the same patient were excluded from the modeling table. This is a deliberate trade-off: it sacrifices some volume in exchange for a dataset in which every row is an independent patient, which is the assumption nearly every standard classifier and evaluation metric depends on. The already-lagged utilization counts (number_outpatient, number_emergency, number_inpatient) are unaffected by this choice, since they describe the year before the retained encounter and require no further adjustment.

Revision to the Original Feature Engineering Plan

The original proposal and EDA Report both anticipated an engineered days_since_prior_encounter feature, measuring the gap between a patient's consecutive visits.

During implementation, this feature was dropped, and that change is documented here rather than silently absorbed into the pipeline, since a stakeholder who read the earlier reports would reasonably expect to see it. The feature was removed for two reasons. First, the only available time reference in this dataset is encounter_id, and the gap between two encounter IDs reflects how many records were logged in the source system between visits, not the number of days that actually elapsed; a difference driven by hospital record volume is not a reliable stand-in for elapsed calendar time, and building a feature on it risked adding noise rather than signal. Second, the feature would have substantially duplicated information the model already receives more directly and more reliably through number_outpatient, number_emergency, and number_inpatient, each of which already summarizes a patient's utilization in the year before the current encounter. Removing a feature that cannot be trusted and is largely redundant with existing predictors is treated here as a methodology improvement, not a loss of scope.

3. Results

3.1 Cleaning, Transformation, and Row-Count Impact

Table 1 traces the dataset's row count through each stage of the cleaning pipeline. Two transformations account for essentially all of the row-count change: removing structurally guaranteed-negative mortality encounters, and collapsing repeated patients down to a single representative encounter. Both were necessary to produce a dataset that is safe to model on, and both trade row count for validity, a trade this project accepts because a larger dataset with a biased or non-independent structure would produce a model whose reported performance could not be trusted in deployment.

Pipeline Stage	Description	Rows Remaining
Raw encounters	Original UCI download, no rows removed	101,766
After mortality exclusion	1,652 in-hospital-death encounters removed (-1.62%)	100,114
After patient-level snapshot	One row retained per patient (earliest eligible encounter); -29.64% vs. prior stage	70,439

Table 1. Row-count impact of the two major cleaning steps.

The target derivation step converted the native three-class readmitted label into the binary readmitted_30d indicator used throughout this report, and then removed the original label from the feature matrix so it cannot leak into the model as a restated version of the outcome it is meant to predict. Standardizing the "?" placeholder to a true null across all columns was completed before any missingness-dependent decision was made, since an accurate missingness count is a prerequisite for every downstream choice described below.

3.2 Missingness Handling Strategy

Table 2 restates the missingness inventory from the EDA Report alongside the concrete handling decision made for each field in this phase. The guiding principle throughout was to match the handling strategy to the mechanism behind the missingness rather than applying one default rule, such as mean imputation or row deletion, across every column, since a single default would have discarded the informative missingness identified in the lab-result fields. Section 2.2 details the imputation alternatives that were explicitly evaluated and rejected before this field-specific strategy was finalized.

Column	% Missing	Handling Strategy	Rationale
weight	96.86%	Dropped entirely	No sample size at which imputation would be reliable
max_glu_serum	94.75%	Binary tested / not-tested indicator	Missingness reflects a clinical testing decision, not a data gap
A1Cresult	83.28%	Binary tested / not-tested indicator	Same as above; testing protocol is itself a signal

Column	% Missing	Handling Strategy	Rationale
medical_specialty	49.08%	Explicit "missing" category	Preserves rows; allows model to learn any administrative pattern
payer_code	39.56%	Explicit "missing" category	Same as above
race	2.23%	Explicit "missing" category	Low missingness; category avoids discarding rows or fabricating a demographic label
admission_type / discharge_disposition / admission_source labels	4.6% to 10.2%	Hidden-unknown placeholder strings consolidated into one "unknown" category	Prevents fragmenting a sparse signal across near-duplicate labels

Table 2. Missingness handling strategy by field, matched to the underlying mechanism.

Table 2b makes the rejected alternatives explicit, field by field, so a reviewer can see not only what was done but what was ruled out and why.

Field	Standard Alternative Considered	Why Rejected
max_glu_serum / A1Cresult	Mean or mode imputation of a lab result; regression / KNN-based imputation	Missingness reflects a physician's decision not to order the test; any filled value fabricates a result that was never recorded
medical_specialty / payer_code	Mode imputation (assign most common category)	High-cardinality, administratively driven fields; mode imputation would overweight the majority category and mask a genuine administrative pattern
race	Mode imputation (assign majority category)	Race is a protected demographic attribute; imputing the majority label would introduce a systematic, direction-biased distortion

Table 2b. Imputation alternatives considered and rejected, by field.

3.3 Diagnosis Code Grouping

Table 3 shows the resulting distribution of the primary diagnosis field (diag_1) after grouping into nine clinical categories, computed on the 100,114 encounters remaining after mortality exclusion. Circulatory conditions are the single largest category by a wide margin, consistent with cardiovascular disease's well-documented prevalence and severity among diabetic inpatients, followed by respiratory and digestive conditions. This distribution will be recomputed for diag_2 and diag_3 using the same mapping logic before modeling begins.

Primary Diagnosis Category	Count	% of Encounters	Clinical Interpretation
Circulatory	29,881	29.85%	Cardiovascular disease is the leading comorbidity in this population
Other / unclassified	17,887	17.87%	Codes outside the nine core chapters (includes V- and E-codes)
Respiratory	14,074	14.06%	Second most common primary diagnosis category
Digestive	9,380	9.37%	Third most common primary diagnosis category
Diabetes (ICD-9 250.x)	8,693	8.68%	Diabetes coded as the primary diagnosis itself, not only a comorbidity
Injury	6,881	6.87%	Trauma-related primary diagnoses
Genitourinary	5,054	5.05%	Kidney and urinary-system conditions, common in advanced diabetes
Musculoskeletal	4,944	4.94%	Bone, joint, and connective-tissue conditions
Neoplasms	3,299	3.30%	Cancer-related primary diagnoses
Missing	21	0.02%	Negligible; consistent with the near-complete diag_1 field noted in the EDA Report

Table 3. Primary diagnosis (diag_1) grouped into nine clinical categories (n = 100,114).

3.4 Patient-Level Deduplication and Time-Aware Train/Test Split

Determining the Split Ratio

Before committing to a specific train and test allocation, an evidence-based split-ratio calculation was run on the patient-level dataset (Step 10) rather than defaulting to a conventional round number. This calculation, which sizes the test set relative to the number of predictors in the feature set, recommended an 87:13 training-to-testing split. That recommendation was not adopted directly. The project's validation strategy requires a chronological split, not a randomly sampled one, because the eventual model will always be scored against patients who come after the ones it was trained on, and a purely proportion-driven calculation has no way to account for that constraint. The two were reconciled by keeping the originally planned 75:25 chronological split: a larger nominal training share is only useful if the holdout it leaves behind still resembles a realistic deployment scenario, and a 13% holdout was judged too thin a slice of the chronology to serve as a trustworthy final evaluation set. This reconciliation, and the reasoning behind it, is reported directly so a reviewer can see that the proportion was checked against evidence rather than assumed, even though the chronological constraint ultimately took priority over the calculated recommendation.

The Primary Split

Table 4 summarizes the patient-snapshot logic described in Section 2.3 and the resulting split. Sorting the post-mortality-exclusion data by the ordinal chronology proxy (encounter_order) and retaining only each patient's first eligible encounter, under the assumption discussed in Section 2.3 that this ordering approximates true calendar time, produced a snapshot of 70,439 unique-patient rows, a 29.64% reduction from the 100,114 encounter-level rows, while lowering the observed 30-day readmission rate to 8.94%. This drop in the positive-class rate is expected under that same assumption and is not, on its own, a data-quality concern: if encounter_order tracks chronology reasonably well, the first-encounter snapshot removes exactly the later, repeat-visit encounters that were disproportionately likely to be readmissions, and it is precisely those repeat encounters that would have violated the independence assumption if left in. The resulting split follows the time-aware, patient-grouped design specified in the proposal: the earliest 75% of the snapshot by the chronology proxy forms the training set, and the most recent 25% is held out for evaluation, with zero patient overlap confirmed between the two partitions. What this section verifies directly is the patient-level independence of the split; what it assumes, and cannot independently verify from the public data, is that "earliest" and "most recent" by encounter_order correspond to genuinely earlier and later calendar time.

Metric	Value	Note
Encounters after mortality exclusion	100,114	Base population for deduplication

Metric	Value	Note
Unique patients in this population	70,439	Each may have 1 or more encounters
Patients with more than one encounter	16,483 (23.4%)	Motivates the snapshot approach
Patient-level snapshot (first eligible encounter)	70,439 rows	-29.64% vs. 100,114 encounter-level rows
Snapshot 30-day readmission rate	8.94%	Overall rate across the full snapshot
Training set (earliest ~75%)	52,829 rows	By ordinal chronology proxy
Training set 30-day readmission rate	9.64%	Computed directly on the training partition
Holdout / test set (most recent ~25%)	17,610 rows	By ordinal chronology proxy
Holdout set 30-day readmission rate	6.82%	Computed directly on the holdout partition
Patient overlap between partitions	0	Verified directly; required for a valid evaluation

Table 4. Patient-level snapshot and time-aware train/test split summary, including the readmission rate observed in each partition.

A second, encounter-level version of this split (75,088 training rows and 25,026 test rows, with the same 75:25 chronological and patient-grouped logic applied) was also built and saved, but it is not used to train the production model. It exists solely as a stretch-goal benchmark, so that later in the project the team can quantify, with real numbers rather than a qualitative guess, how much predictive performance and feature stability change when repeated patient encounters are treated as independent rows instead of collapsed to one snapshot per patient. Any result drawn from that encounter-level version should be read as a secondary, qualitative comparison, since it still contains the non-independence problem described in Section 2.1.

This benchmark also produces a useful, if secondary, piece of evidence for the split-balance question addressed directly below. On the encounter-level split, the training and holdout readmission rates are 11.23% and 11.68%, a gap of only about 0.45 percentage points, versus the 2.8-point gap observed on the primary patient-level split. That the encounter-level split is comparatively balanced while the patient-level split is not is consistent with the explanation given below: the gap is tied to which encounters the patient-level snapshot keeps and discards (each patient's first eligible encounter, disproportionately from earlier in the chronology) rather than being an artifact of the time-aware cut point itself, since that same cut point produces a much smaller gap when applied to the full encounter-level data.

Verifying the Split's Impact on the Outcome

The training and holdout partitions do not carry the same readmission rate: 9.64% in training versus 6.82% in the holdout set, a gap of roughly 2.8 percentage points. This gap is reported directly here rather than left for a reader to infer, because a time-aware split is expected to behave differently from a random split on exactly this dimension, and a stakeholder evaluating the eventual model's holdout metrics needs to know the two partitions are not drawn from an identical outcome distribution.

Three explanations were considered, and they are not mutually exclusive: a structural artifact of the time-aware cut point itself, a genuine shift in patient mix or hospital discharge practice over the dataset's 1999 to 2008 span, or a breakdown in the chronology-proxy assumption itself, meaning encounter_order does not track calendar time closely enough for the split to be genuinely time-ordered. The encounter-level benchmark reported just above provides partial evidence on the first explanation. Because that benchmark applies the same chronological cut point to the full, non-deduplicated data and produces a much smaller gap (0.45 points versus 2.8 points), the time-aware cut point on its own does not appear to be the primary driver. The more likely explanation is tied specifically to the patient-level snapshot: retaining only each patient's first eligible encounter concentrates the training partition with earlier-chronology, first-time encounters and the holdout partition with a comparatively higher share of first-time encounters that happen to fall later in the sequence, which is not equivalent to a uniform shift in the underlying population. This does not fully rule out a genuine temporal trend in discharge practice, and it does not rule out the third explanation either: because the entire earlier-versus-later framing depends on encounter_order approximating true chronology, a gap this size is also consistent with encounter_order being a noisier proxy than assumed, in which case some of what looks like a genuine 1999-to-2008 shift could instead be an artifact of imperfect ordering. This report does not have the evidence to rule that possibility out, and says so plainly here rather than defaulting to the more comfortable explanation. What can be said with more confidence is that the gap should be treated as tied to the snapshot design choice made in Section 2.3 rather than assumed away, and the gap itself changes how the modeling phase should proceed in three concrete ways.

- **Do not benchmark holdout recall or precision against the 8.94% snapshot-wide base rate.** The relevant reference point for interpreting holdout performance is the holdout set's own 6.82% rate, not the pooled figure, or metrics will be misread as better or worse calibrated than they are.
- **Report cross-validation folds within the training partition only, as already planned,** but additionally check whether readmission rate drifts across folds within training. If it does, that is further evidence for a temporal trend rather than a one-time artifact of the train/holdout cut point.
- **Treat the holdout evaluation as a conservative, not an optimistic, test.** Because the positive class is rarer in the holdout partition than in training, a model calibrated on training-set prevalence may need its decision threshold re-tuned rather than applied unchanged, a step that is now added explicitly to the modeling plan in Section 4.2. This report frames the gap as conservative because a lower holdout base rate typically makes

precision look worse and recall look no easier to achieve than the training figures would suggest, not because the direction of the gap is confirmed to be benign. Three distinct explanations remain open, are not mutually exclusive, and are not fully distinguished by the evidence gathered so far: the snapshot rule itself, retaining each patient's earliest eligible encounter, concentrating early-chronology rows into training; a genuine shift in patient mix or hospital discharge practice across the dataset's 1999 to 2008 span; or a breakdown in the chronology-proxy assumption described in Section 2.3, where encounter_order fails to track true calendar time closely enough for "earliest" and "latest" to mean what they are assumed to mean. Distinguishing between these would require evidence this phase does not yet have, for example checking whether the gap persists when the last, rather than the first, eligible encounter is kept per patient (the sensitivity variant already built and saved alongside the primary snapshot), or examining whether the readmission rate drifts smoothly across the training partition's own chronology rather than jumping only at the train/holdout boundary. Until one of those checks is run, the safest posture is to treat the reported holdout metrics as informative but not fully explained, and to flag the open question in the Limitations discussion in Section 4.3 rather than resolve it by assumption.

3.5 Evaluation of Unsupervised and Supervised Feature Engineering Directions

The original proposal called for two unsupervised techniques, PCA and k-means clustering, as part of the exploratory analysis plan. Both were evaluated using the same training partition and preprocessing pattern established in this phase, and deliberately excluded from the deployment pipeline, for reasons stated directly below with supporting quantitative evidence rather than left implicit, since a stakeholder reviewing this pipeline should know that these techniques were tried and rejected on the basis of specific diagnostics, not simply skipped or rejected on intuition alone.

Technique	Evaluation Finding	Decision & Rationale
Principal Component Analysis (PCA)	PCA was fit on the full 243-column one-hot-encoded training matrix (49 predictors before encoding). The leading component (PC1) explains only 10.4% of total variance, and the first 20 components together reach just 73.9% of cumulative variance; a full 26 components, out of 243, are needed to reach the conventional 80% threshold (Table 5b). This confirms quantitatively, rather than by assumption, that variance is spread thinly across many sparse one-hot dimensions rather than concentrated in a small number of components. The strongest correlation between any of the first 10 components and the readmission target is 0.071 (PC1), too weak to justify treating a component as a useful engineered feature. The resulting components would also combine multiple raw fields into linear blends that a care manager cannot map back to a specific clinical action.	Excluded from the deployment pipeline. Interpretability is a hard requirement for this project's stakeholder audience, and the variance and target-correlation results above show PCA trades that requirement away without a proportionate reduction-in-dimensionality benefit.
K-means clustering	K-means was run on eight utilization and treatment-intensity fields plus age bracket and primary diagnosis group, swept across k = 2 through 8 (Table 5b). Silhouette scores were low across every candidate k (0.12 to 0.18 on a scale where scores near 1.0 indicate well-separated clusters), and never approached a range that would indicate distinct, well-separated groupings. k = 4 was selected using the elbow method on inertia; the resulting clusters range from 2,860 to 21,127 patients, and one small cluster (n = 2,860) does show a notably elevated readmission rate of 19.6% against an overall training rate of 9.6%, which is a potentially useful care-management signal even though the clusters overall are not well separated.	Excluded from the automated deployment pipeline as a modeling input, on the strength of the uniformly low silhouette scores above; a tree-based model's own decision splits can capture comparable structure without the added complexity of a separate clustering step. The one high-risk cluster identified is flagged in Section 4.2 as a candidate for care-management targeting outside the supervised model, since that finding does not depend on the clusters being well separated overall.

Table 5. Unsupervised techniques evaluated and excluded, with rationale.

Table 5b makes the underlying diagnostics explicit, since a rejection decision is only as credible as the evidence behind it. Both analyses were run on the training partition of the patient-level dataset established in this phase, using the training-only, leakage-safe preprocessing pipeline; the scree plot and elbow/silhouette curves referenced below are retained in the project repository alongside the code that produced them.

Diagnostic	Value	Interpretation
Clustering variables	8 utilization/treatment-intensity fields (time_in_hospital, num_medications, num_lab_procedures, num_procedures, number_outpatient, number_emergency, number_inpatient, number_diagnoses) + age bracket + primary diagnosis group	Chosen for clinical interpretability rather than the full one-hot modeling matrix
Candidate k values swept	k = 2 through 8	Elbow (inertia) and sampled silhouette score computed at each k
Silhouette score range across k	0.117 to 0.180 (best at k = 2: 0.180)	All values are low; none approach the 0.5+ range associated with well-separated clusters
Selected k (elbow method)	k = 4	Cluster sizes range from 2,860 to 21,127 patients; silhouette at k = 4 is 0.117, among the lowest of the candidates tested
Highest-risk cluster readmission rate	19.58% (n = 2,860) vs. 9.64% training-set average	Flagged separately in Section 4.2 as a care-management targeting signal, independent of overall cluster separability
PCA components fit	20 (of 243 one-hot-encoded columns; 49 predictors before encoding)	Fit on the training partition only, using the same preprocessing pipeline as the supervised model
PC1 explained variance	10.4%	No single component captures a large share of variance
Cumulative variance at 20 components	73.9%	Well short of the conventional 80% reference line
Components needed for 80% cumulative variance	26 of 243	Confirms variance is spread thinly across the sparse feature space rather than concentrated
Strongest PC-target correlation (PC1-PC10)	0.071 (PC1)	Too weak to justify using a principal component as an engineered feature

Table 5b. Quantitative diagnostics supporting the PCA and k-means exclusion decisions in Table 5.

Excluding these techniques from the deployed pipeline is a scope decision, not an abandonment of the original analysis plan; both were run and documented, and both are retained in the project's GitHub repository as evaluated-and-rejected approaches so the reasoning is auditable.

Supervised feature engineering will proceed instead along three lines during the modeling phase, using train-set-only transformations to avoid leakage: target encoding (smoothed with a global prior probability, fit only within cross-validation folds) for the high-cardinality medical_specialty field; interaction terms between prior-utilization flags and age bracket, to capture compounding risk among older patients with frequent recent hospitalizations; and gradient-boosted feature importance (gain and permutation importance) to prune redundant features before a model is promoted to production.

4. Discussion & Next Steps

4.1 Summary of Key Takeaways

This phase converted the raw encounter-level extract into a leak-free, patient-independent, model-ready table, and every transformation applied is traceable to a specific finding from the EDA Report rather than a default assumption. Four points are most consequential for the modeling phase that follows.

- **The data continues to support the project's hypothesis.** Even after removing mortality-biased encounters and collapsing to one row per patient, prior-utilization and treatment-intensity fields remain distributionally distinct between readmitted and non-readmitted patients, consistent with the pattern first observed in the EDA Report.
- **The patient-level snapshot is a deliberate, disclosed trade-off.** Reducing the dataset from 100,114 encounters to 70,439 independent patient rows lowers the observed readmission rate to 8.94% and further shrinks an already modest positive class. This makes the class-imbalance handling specified in the original proposal (class weighting as

the primary approach, with SMOTE evaluated only as a comparison) more important in the modeling phase, not less.

- **The time-aware, patient-grouped split is implemented and verified, not just planned.** With zero confirmed patient overlap between the training and holdout partitions, the split is structurally sound. However, verifying the split also surfaced a 2.8-percentage-point gap in readmission rate between training (9.64%) and holdout (6.82%), which is addressed directly in Section 3.4 and now carries forward as an explicit threshold-tuning step in the modeling plan below, rather than an assumption that holdout performance can be read at face value.
- **Imputation was deliberately not used as a default missing-data strategy.** Mean, median, and mode imputation were each evaluated against the specific mechanism behind this dataset's nine incomplete fields and rejected in every case in favor of indicator flags or explicit missing categories, documented field by field in Section 3.2, so that no missing value in this dataset was silently fabricated.

4.2 Modeling Plan and Next Steps

With clean, split, and leak-checked data in hand, the project moves into supervised modeling. The plan below follows directly from the validation strategy established in the proposal and confirmed in this phase.

- Establish cross-validation pipelines using only the training partition (52,829 rows), so that no evaluation decision is made using information from the holdout set.

- Train baseline models, a dummy majority-class classifier and a regularized logistic regression, to set the minimum bar for AUC-ROC and recall that any more complex model must clear to justify its added complexity.

- Train non-linear supervised models, specifically random forest, LightGBM, and XGBoost, to capture the utilization and diagnosis-driven interactions the EDA Report flagged as unlikely to be captured by a purely linear baseline.

- Apply class weighting to address the 9.64% training-set positive-class rate, and re-tune the decision threshold specifically against holdout-set prevalence (6.82%) rather than the training rate, consistent with the split-imbalance finding in Section 3.4, to maximize recall for high-risk patients while keeping the false-alert rate at a level care teams can realistically act on.

- Compute SHAP values for the leading candidate model, both to confirm whether utilization-related features rank among the top predictors, as the hypothesis predicts, and to give care management stakeholders an interpretable basis for trusting individual risk scores.

- Promote the best-performing model into the MLflow model registry and serve it through a FastAPI endpoint, completing the Phase 1 deliverable described in the original proposal's timeline.

4.3 Limitations and Assumptions Carried Into the Modeling Phase

The preceding sections state individual assumptions where they arise. This section collects the two most consequential ones in one place, on request, because both the first-encounter snapshot in Section 2.3 and the chronological train/test split in Section 3.4 depend on them, and a stakeholder evaluating this project's holdout results should be able to see the full extent of that dependence without hunting for it across the report.

- **Encounter_id as a chronology proxy is an assumption, not a verified fact.** This report assumes that sorting by encounter_id reproduces the true calendar order in which encounters occurred. The only support for that assumption is the UCI data dictionary's description of encounter_id as a sequentially assigned identifier; no calendar timestamp exists in the public release to confirm it directly, so the assumption has not been, and cannot be, independently tested in this phase. If the assumption is wrong, meaning encounter IDs were assigned in a way that only loosely tracks calendar time, for example through batch loading, backfilling, or merging records from multiple hospitals out of strict time order, then two downstream results are affected. First, the first-encounter snapshot may not reliably capture each patient's earliest visit, which would weaken the argument that later, more readmission-prone encounters were cleanly excluded. Second, and more seriously, the train/test split may not be genuinely time-ordered, which would undercut the central justification for using a chronological split at all: evaluating the model on data that plausibly resembles what it will see in deployment, after the data it was trained on. This report proceeds with the assumption because it is the only ordering signal the public data provides and because the original proposal committed to a time-aware validation strategy, not because the risk has been ruled out.

- **The 2.8-point readmission-rate gap between training and holdout is not fully explained.** As discussed in Section 3.4, this gap is consistent with at least three explanations: the patient-level snapshot design, a genuine shift in patient mix or discharge practice over the dataset's 1999 to 2008 span, or a breakdown in the chronology-proxy assumption above. The evidence gathered in this phase narrows the field somewhat, the encounter-level benchmark suggests the time-aware cut point alone is not the primary driver, but it does not fully distinguish between the remaining explanations. Until a first-versus-last-encounter sensitivity check or a within-training drift check is run, holdout metrics in the modeling phase should be interpreted with this open question in mind rather than treated as a fully understood baseline.

Neither limitation is a defect introduced by an error in this phase; both are structural properties of a public dataset that does not include calendar timestamps, combined with modeling choices, the patient-level snapshot and the chronological split, that this report continues to judge as the right choices despite the uncertainty they carry. The purpose of stating them here explicitly is so that any conclusion drawn from the holdout set in the modeling phase can be weighed against the specific assumptions it rests on, rather than read as unconditionally reliable.

Works Cited

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Appendix. Data Dictionary and Codebook

The table below lists every variable that exists in the model-ready table produced by this phase, following the same format as the data dictionary in the EDA Report's appendix. Engineered variables are marked as such; they do not exist in the raw UCI download and were constructed during preprocessing or feature engineering as described in Sections 2 and 3.

Variable	Data Type	Source / Status	Description
patient_nbr	Integer (ID)	Source	Unique patient identifier; used as the deduplication and grouping key for the patient-level snapshot.
encounter_id	Integer (ID)	Source	Unique identifier for a single inpatient encounter; underlies the ordinal chronology proxy.
encounter_order	Integer	Engineered	Sequential rank derived from encounter_id, used as a stand-in for calendar time in the split.
readmitted_30d	Binary (0/1)	Engineered / TARGET	Collapsed from the native three-class label; 1 if readmitted within 30 days.
age	Categorical	Source	Ten-year age bracket.
race	Categorical	Standardized	Includes an explicit missing category for unrecorded values.
gender	Categorical	Source	Patient gender classification.
admission_type_id_label	Categorical	Mapped / clean	Descriptive admission type; hidden-unknown placeholders consolidated.
discharge_disposition_id_label	Categorical	Mapped / clean	Descriptive discharge destination; hidden-unknown placeholders consolidated.
admission_source_id_label	Categorical	Mapped / clean	Descriptive referral source; hidden-unknown placeholders consolidated.
time_in_hospital	Integer	Source	Length of stay in days.
num_lab_procedures	Integer	Source	Number of lab tests performed during the encounter.
num_procedures	Integer	Source	Number of non-lab procedures performed during the encounter.

Variable	Data Type	Source / Status	Description
num_medications	Integer	Source	Number of distinct medications administered during the encounter.
number_outpatient	Integer	Source	Outpatient visits in the year prior to the encounter (already lagged).
number_emergency	Integer	Source	Emergency visits in the year prior to the encounter (already lagged).
number_inpatient	Integer	Source	Inpatient admissions in the year prior to the encounter (already lagged).
diag_1_group / diag_2_group / diag_3_group	Categorical	Engineered	Primary, secondary, and tertiary diagnoses grouped into 9 clinical categories.
number_diagnoses	Integer	Source	Total number of diagnoses recorded for the encounter.
max_glu_serum_tested	Binary categorical	Engineered	Whether a glucose serum test was ordered (Tested / Not Tested).
A1Cresult_tested	Binary categorical	Engineered	Whether an HbA1c test was ordered (Tested / Not Tested).
change	Categorical	Source	Whether diabetic medication dosage was altered during the visit.
diabetesMed	Categorical	Source	Whether any diabetic medication was prescribed.

Table A1. Data dictionary for the model-ready table produced by this phase.